A matrix $M(t)$ is said to be positive (semi-) definite if

$$x^T M(t) x > 0 \ (\geq 0) \quad \forall t$$

$\Rightarrow$ (By Rayleigh-Ritz)

$$\min_t \lambda_{min}(M(t)) > 0 \ (\geq 0)$$

$M(t)$ is <u>uniformly</u> PD if

$$\min_t \lambda_{min}[M(t)] \geq \eta > 0 \quad \forall t.$$

Previous Result, re-interpreted

We have shown $\frac{d}{dt}\|x\|_2^2 = -x^T Q(t) x$

where $Q(t) = -[A^T(t) + A(t)] = -2A_{sym}(t)$

(recall $A_{sym} = \frac{1}{2}(A + A^T)$)

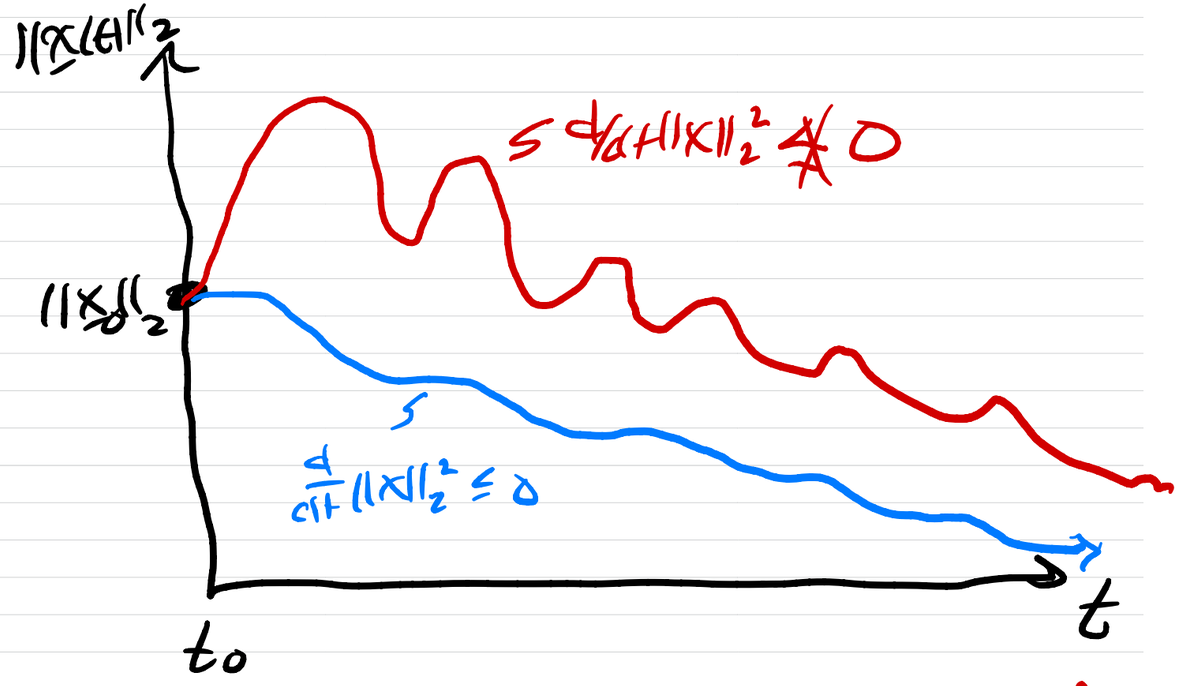
So that $\frac{d}{dt}\|x\|_2^2 \leq 0$ if

$$\min_t \lambda_{min}(Q(t)) \geq 0$$

i.e. if $Q(t)$ is pos semi-definite. Equivalently

$$\max_t \lambda_{max}[A^T(t) + A(t)] \leq 0$$

Note $\frac{d}{dt}\|x\|_2^2 \leq 0$ is too restrictive to fully characterize stability



Red is also stable, but doesn't satisfy $\frac{d}{dt}\|x\|_2^2 \leq 0 \quad \forall t \geq t_0$

Reinterpret the argument more geometrically.

Let $V(\underline{x}) = \|\underline{x}\|_2^2$

Note that:

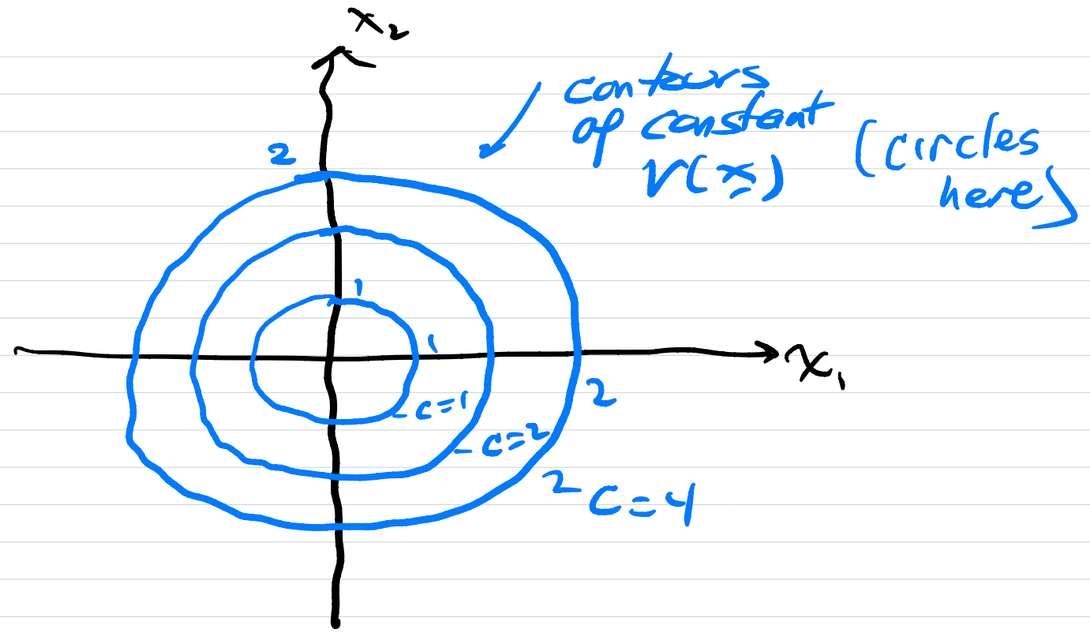
1.) The "level sets" $V(\underline{x}) = c$ for $c \geq 0$ define a nested family of closed contours.

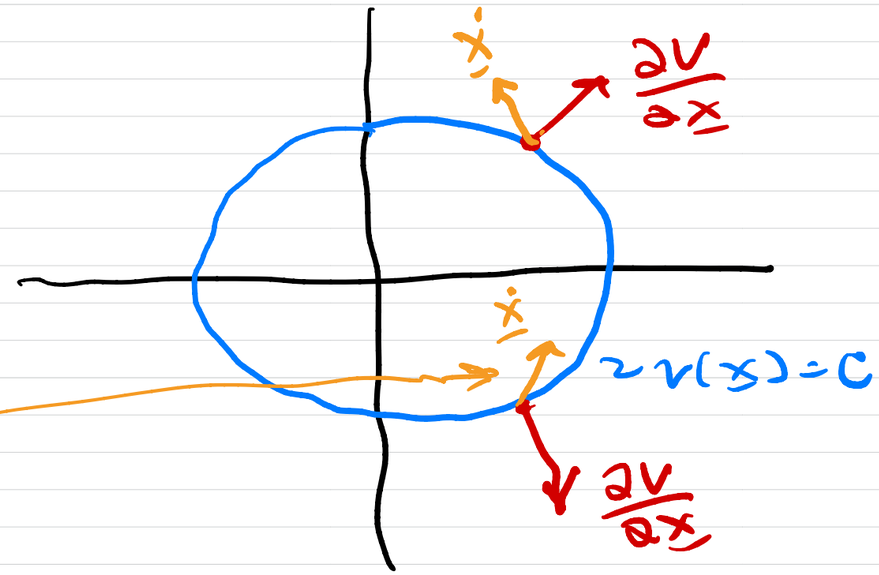
2.) $\frac{d}{dt}\|\underline{x}\|_2^2 = \frac{d}{dt} V(\underline{x})$

$$= \frac{\partial V}{\partial \underline{x}} \cdot \dot{\underline{x}}$$

$\frac{\partial V}{\partial \underline{x}}$: outward Normal vector to contour lines

$\dot{\underline{x}}$: Direction state vector is moving





$\dot{V} \leq 0$

$$\Rightarrow \frac{dV}{\partial \underline{x}} \cdot \dot{\underline{x}} \leq 0$$

$\Rightarrow \dot{\underline{x}}$ points into the contour (or at worst along the bdy)

$$\frac{\partial V}{\partial \underline{x}} \cdot \dot{\underline{x}} \leq 0 \quad \forall \underline{x}$$

$\Rightarrow$ motion at any point on any contour is directed to interior of that contour

This means that the interior of each contour:

$$S(c) = \{x \mid V(x) \leq c\}$$

defines an "invariant set", i.e.

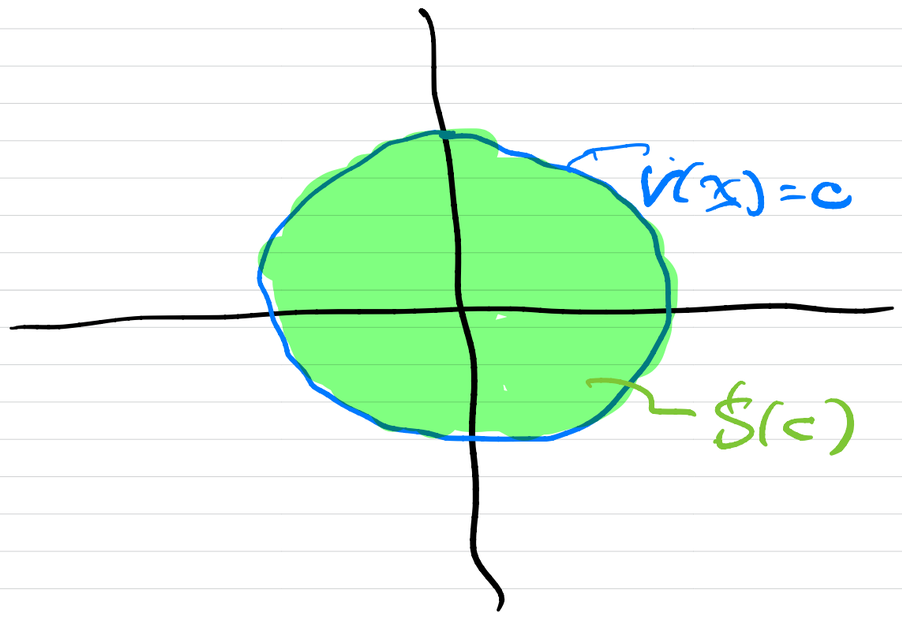
if $\exists t_1$ s.t. $x(t_1) \in S(c)$ then necessarily $x(t) \in S(c)$ $\forall t \geq t_1$

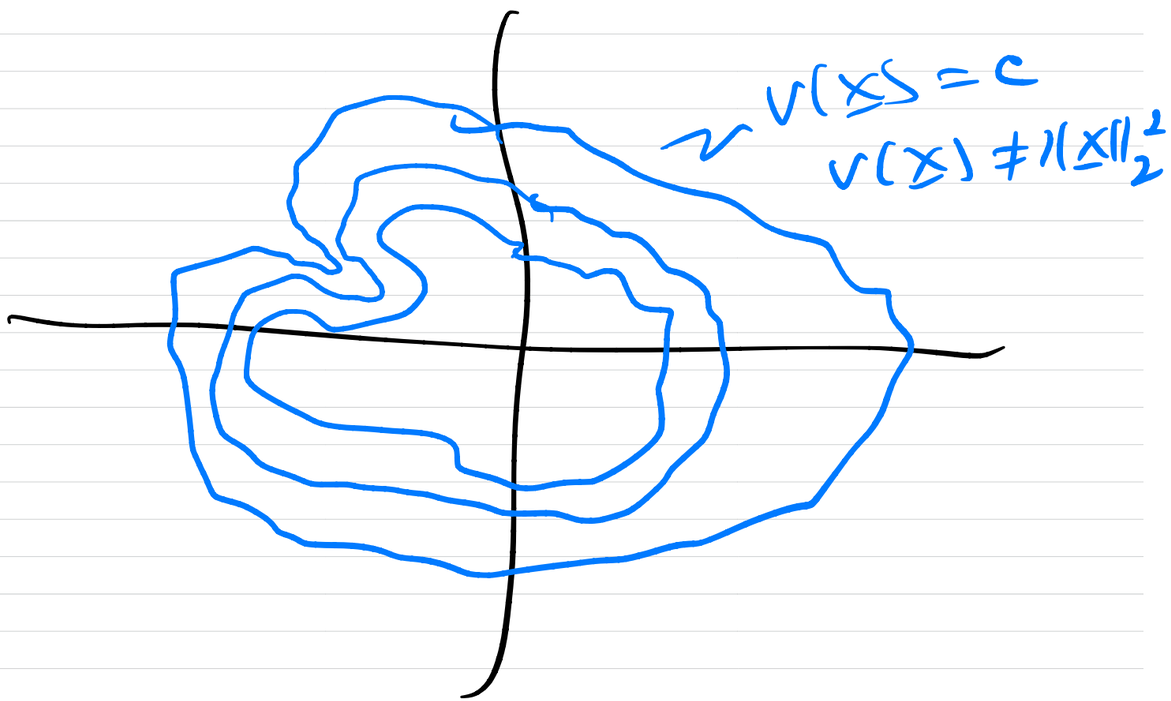
$\frac{\partial V}{\partial x} \cdot \dot{x} \leq 0$ $\forall x$ means

each contour is an invariant set, and state can never "escape" the contour it starts inside $\Rightarrow$ stability ish

= Note: nothing in these arguments requires circular contours!

Any nested set of closed contours works

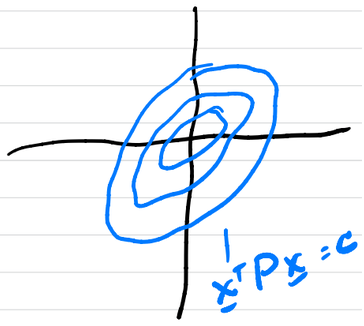




For Linear systems, we need only consider $V(\underline{x})$ for which the level sets $V(\underline{x}) = c$ are elliptical which can be described using the quadratic form

$$V(\underline{x}) = \underline{x}^T P \underline{x}$$

P positive definite.



=

Terminology:

General $V(\underline{x})$ satisfying the Nested contour requirement above is known as a "Lyapunov function". A quadratic form for $V(\underline{x})$ is known as a quadratic Lyapunov f'n (QLF)

Note that our original

$$V(\underline{x}) = \|\underline{x}\|_2^2$$

$$= \underline{x}^T P \underline{x}$$

with $P = I$

Suppose matrix $P(t)$ is uniformly bounded and uniformly positive definite, i.e.

$$\rho = \max_t \lambda_{max}(P(t)) < \infty$$

$$\zeta = \min_t \lambda_{min}(P(t)) > 0$$

Define

$$-Q(t) = \dot{P}(t) + A^T(t)P(t) + P(t)A(t)$$

Lyapunov Eq'n.

$$-\dot{Q}(t) = \dot{P}(t) + A^T(t)P(t) + P(t)A(t)$$

Then

1.) If $Q(t)$ is positive semi-definite, then system is stable i.S.L.

2.) If $Q(t)$ is uniformly positive definite, then system is exponentially stable.

Pf:

Let $V(x,t) = x^T P(t) x$

$$\dot{V} = \frac{d}{dt} V(x(t),t)$$

$$= \dot{x}^T P x + x^T \dot{P} x + x^T P \dot{x}$$

$$= (Ax)^T P x + x^T \dot{P} x + x^T P (Ax)$$

$$= x^T [\dot{P} + A^T P + PA] x$$

$$= -x^T Q(t) x$$

Note: if tests fail the result is inconclusive → a different choice of $P(t)$ may yield different results.

⇒

Is it true that every stable LES system will have (P, Q) satisfying required cond'ns?

YES! In fact infinitely many P, Q pairs can be found.

Indeed

$$P(t) \triangleq \int_t^\infty \Phi^T(\tau,t) Q(\tau) \Phi(\tau,t) d\tau$$

↳ STM for $A(t)$

satisfies Lyap eqn.

1.) Q pos semi-def $\Rightarrow \dot{V} \leq 0$

$$\gamma \|\underline{x}(t)\|_2^2 \leq \underline{x}^T(t) P(t) \underline{x}(t) = V(\underline{x}(t), t) \leq V(\underline{x}_0, t_0) = \underline{x}_0^T P(t_0) \underline{x}_0 \leq \rho \|\underline{x}_0\|_2^2$$

Thus

$$\gamma \|\underline{x}(t)\|_2^2 \leq \rho \|\underline{x}_0\|_2^2$$

$$\Rightarrow \|\underline{x}(t)\|_2 \leq \sqrt{\frac{\rho}{\gamma}} \|\underline{x}_0\|_2$$

Def'n of ish stability.

2.) Q uniformly positive definite

$$\Rightarrow \omega = \min_t \lambda_{min}(Q(t)) > 0$$

Then

$$\dot{V} = -\underline{x}^T Q(t) \underline{x}$$

$$\leq -\omega \|\underline{x}(t)\|_2^2$$

To make further progress we need an important lemma.

Lemma ("Comparison lemma")

Suppose $\dot{\omega}(t) = f(\omega(t), t) \quad \omega(t_0) = \omega_0$

has known sol'n $\omega(t)$

Then Differential inequality

$$\dot{v}(t) \leq f(v(t), t) \quad v(t_0) = v_0 \leq \omega_0$$

has sol'n $v(t)$ satisfying

$$v(t) \leq \omega(t) \quad \forall t \geq t_0$$

$$\dot{V} = -\underline{x}^T Q(t) \underline{x}$$
$$\leq -\nu \|\underline{x}\|_2^2$$

Not that $\underline{x}^T P \underline{x} \leq \rho \|\underline{x}\|_2^2$ — V

$$\Rightarrow -\|\underline{x}\|_2^2 \leq (-1/\rho)\, \underline{x}^T P \underline{x}$$

Hence

$$\dot{V} \leq -\nu \|\underline{x}\|_2^2$$
$$\leq -(\nu/\rho)\, \underline{x}^T P \underline{x} \quad V$$

OR

$$\boxed{\dot{V} \leq -(\nu/\rho)\, V}$$

By comparison lemma:

$$\Rightarrow V(t) \leq V(t_0) e^{-\lambda_1 (t - t_0)}$$

with $\lambda_1 = \nu/\rho$

Aside

Suppose

$$\dot{V} = \alpha(t)\, V(t)$$

if $\alpha(t) > 0 \ \forall t$

does $V(t) \to \infty$ necessarily?

So: $V(t) \le V(t_0) e^{-\lambda_1 (t-t_0)}$

with $\lambda_1 = \gamma/\rho$

$\eta \|\underline{x}(t)\|_2^2 \le \underline{x}^T(t) P(t) \underline{x}(t) = V(t) \le V(t_0) e^{-\lambda_1 (t-t_0)} = \underline{x}_0^T P(t_0) \underline{x}_0 e^{-\lambda_1 (t-t_0)} \le \rho \|\underline{x}_0\|_2^2 e^{-\lambda_1 (t-t_0)}$

$$\Rightarrow \quad \eta \|\underline{x}(t)\|_2^2 \le \rho \|\underline{x}_0\|_2^2 e^{-\lambda_1 (t-t_0)}$$

$$\Rightarrow \quad \|\underline{x}(t)\|_2 \le \sqrt{\frac{\rho}{\eta}}\, e^{-\lambda (t-t_0)} \|\underline{x}_0\|_2$$

with $\lambda = \frac{\gamma}{2\rho}$

Note λ is determined by ratio of min eigenvalue of $Q^{-(p)}$ to max eigenvalue of $P^{-(p)}$